AMEYA PARAB

PhD Research Scholar

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Personal Statement

Laser-plasma researcher pursuing an integrated PhD at the Tata Institute of Fundamental Research (TIFR), with a background in mechanical engineering. Experienced in high-intensity femtosecond laser experiments, plasma diagnostics, and particle-in-cell simulations. Research focuses on electron acceleration, magnetic-field generation, and shock-wave dynamics in laser-produced plasmas. Combines experimental investigation and numerical modelling, with contributions to laser experiments at TIFR and ELI-NP, Romania, and the development of experimental setups and diagnostic systems.

Qualifications

Integrated PhD **2021- Present**

TIFR (Tata Institute of Fundamental Research)
Department of Atomic and Nuclear Physics

Bachelors of Engineering (Mechanical Engineering) **2015 - 2019**

Mumbai University
Bharati Vidyapeeth College of Engineering

Work Experience

PhD Thesis

- Conducted an experiment at the 1 PW laser facility at ELI-NP, Romania, to investigate directional MeV gamma-ray emission alongside electron, proton, and neutron emission.
- Developed a framework for direct laser acceleration in the presence of longitudinal and transverse electric fields, and proposed a practical multi-beam electron-acceleration scheme using two-dimensional particle-in-cell simulations with SMILEI on the TIFR HPC cluster.
- Investigating shock waves generated in plasma by high-contrast, 400 nm laser pulses and studying how laser absorption varies with pulse energy, angle of incidence, and externally applied prepulse conditions.

Contributing Projects

- Setting up a Stokes polarimetry system to measure megagauss magnetic fields using pump-probe diagnostics in an intense, ultrashort, terawatt laser system.
- Participated in experiments investigating shock propagation through Doppler-shift measurements and wavefront sensing using high-contrast, ultrashort second-harmonic pump and probe pulses.
- Contributed to planning and setting up experimental chmabers and diagnostics at the new petawatt laser facility in Hyderabad, India.

Work Experience (Continued)

Engineering Projects

- Contributed to planning and building a Formula Student car and led the steering team for Formula Bharat 2019.
- As a final year project, developed a self-adjusting monitor as a final-year engineering project, using face recognition and a gear assembly to maintain a minimum viewing distance and support improved posture.

Presentations, Conferences & Training

- Summer School (LaPlass 2023) CLPU Salamanca, Spain.
- Strong Plasma Radiation Interaction Group - II at IIT Delhi.
- Nuclear Photonics confernece at TIFR, Hyderabad.
- Winter School on Intense Science (WISILS) at IIT Jodhpur, India.
- Poster presented at Conference on Plasma Simulation (CPS2024) at Indian institute of Geomagnetism, Mumbai.
- Poster presentation, 13th Asian Symposium on intense laser science (ASILS13) at Udaipur, India.
- Oral Preseantation at National Symposium on Intense Laser Matter Interactions 2026 (ILMI 2026) in IIT (ISM) Dhanbad.
- Oral Presentation at PSSI- PSC 2025 held at IPS Acadamy Indore.
- Oral Presentation at Frontier in Optics + Laser Science (FIOLS) 2026 held in Rochester, USA.
- Participated in Student leadership programme in Rochester, USA.

Software Skills

SMILEI (PIC Simulaton)	EPOCH (PIC Simulation)	LabVIEW
LaTeX	Python	MATLAB
AutoCAD	Origin	Wolfrom Mathematica
SolidWorks	Microsoft Excel	Microsoft PowerPoint
Microsoft Word	Adobe Photoshop	Adode Lightrom

Technical Skills

Leadership	Problem Solving	Imterpersonal Skills
Communication	Adaptibility	Critical Thinking
Time Management	Work Ethic	Creativity
Teamwork	Conflict Resolution	Emotional Intelligance

Languages

English (Fluent)	Marathi (Native)	Hindi

Publications

- **“Bright, directional electron emission from nanowire coated targets under petawatt, femtosecond irradiation”**

  https://doi.org/10.1088/1367-2630/aea590

  Ameya Parab, Jian Fuh Ong, Stefania Ionescu, Sagar Dam, Sk Rakeeb, Hideaki Habara, Y.Keita, Rudrajyoti Palit, Daniel Popa, Deepak Sangwan, Klaus Spohr, Lucian Tudor, Adrian Vatcu, Prashant Kumar Singh, Kazuo A. Tanaka, G.Ravindra Kumar.

- **“Quasi-static transverse electric field driven electron acceleration in relativistic laser matter interaction”**

  https://doi.org/10.48550/arXiv.2607.26252

  Ameya Parab, Bhooshan Paradkar, Aparajit C., Anandam, Sk Rakeeb, Sagar Dam, Prashant Kumar Singh

- **“Tens of MeV, collimated, bright fluxes of protons from ordered nano- structured targets in ultra-relativistic laser-matter interaction”**

  https://doi.org/10.48550/arXiv.2607.09229

  Sagar Dam, Stefania Ionescu, Jian Fuh Ong, Ameya Parab, Sk Rakeeb, Hideaki Habara, Gabriel Cojocaru, Vojtěch Horný, Dmitrii Nistor, Saidbek Norbaev, Rudrajyoti Palit, Daniel Popa, Deepak Sangwan, Klaus Spohr, Bianca Stan, Antonia Toma, Lucian Tudor,

- **“Impulsive excitation of a solid by extreme contrast, high- intensity femtosecond laser pulses”**

  https://doi.org/10.1103/dtc5-ftnd

  Sagar Dam, J. F. Ong, Sk Rakeeb, Ameya Parab, Aparajit C, Anandam, Amit D. Lad, Yash M. Ved, M. Krishnamurthy, K. A. Tanaka and G. Ravindra Kumar.

- **“Capturing the three dimensional, nanoscale, picosecond dynamics of plasma mirrors with intense ultrashort laser wavefront measurement”**

  https://doi.org/10.1364/OL.570014

  Sk Rakeeb, Sagar Dam, Ameya Parab, Amit D. Lad, Yash M. Ved, and G. Ravindra Kumar.

- **“Spatial, Spectral and Temporal Response of High Intensity Laser Plasma Mirrors- Direct Observation of the Ponderomoti ve Push”**

  https://doi.org/10.48550/arXiv.2506.18425

  Sk Rakeeb, Animesh Sharma, Sagar Dam, Ameya Parab, Amit Lad, Yash Ved, Amita Das, G Ravindra Kumar

- **“Generation of mega-gauss axial and azimuthal magnetic fields in a solid plasma by ultrahigh intensity, circularly polarized femtosecond laser pulses”**

  https://doi.org/10.48550/arXiv.2405.14375

  Anandam Choudhary, Laxman Prasad Goswami, C Aparajit, Amit D Lad, Ameya Parab, Yash M Ved, Trishul Dhalia, Amita Das, G Ravindra Kumar

- **“10^4- fold amplification of a tiny magnetic field to megagauss scale in femtosecond , ulraintense laser-solid interaction”**

  https://doi.org/10.48550/arXiv.2504.15094

  Anandam Choudhary, Trishul Dhalia, Sagar Dam, Ameya Parab, SK Rakeeb, C Aparajit, Amit D Lad, Yash M Ved, Kandaswamy Subramanian, Amita Das, G Ravindra Kumar